

SETU: Primary Mobile App & On-Device AI Architecture

Technical Specification: Edge Sensor Ingestion, DSP Math, Feature Extraction & Cloud Payloads

Architectural Role (Layer 1 Edge SDK): The SETU primary application runs as an ultra-lightweight (~200 KB SDK) inside gig-worker host apps (Swiggy, Zomato, Ola, Rapido). It utilizes a strictly *edge-first, battery-protective* design: deterministic Digital Signal Processing (DSP) and hard statistical gating eliminate **~99% of regular driving windows** on the CPU. A quantized 120 KB LiteRT/TFLite 1D-CNN runs only on candidate shock events to classify road defects, uploading rich metadata every 60 seconds while keeping battery drain < 2%/hr and mobile data < 1 MB/day.

1. Direct Hardware & Sensor Measurements (Sampled Continuously)

The app interfaces directly with device hardware via Android SensorManager and FusedLocationProviderClient:

Hardware Channel	Sampling Rate	Physics & Purpose Measured
3-Axis Accelerometer (aX, aY, aZ)	100 Hz (~10 ms)	Measures linear acceleration and kinetic impact forces across X, Y, and Z axes.
3-Axis Gyroscope (gX, gY, gZ)	100 Hz (~10 ms)	Measures angular velocity (roll, pitch, yaw rates); detects single-wheel roll asymmetry when hitting a pothole.
Gravity Vector (gx, gy, gz)	100 Hz / Continuous	Measures static gravitational field direction to calculate real-world 'down' and device orientation.
GPS Coordinates (Lat, Lon)	1 Hz / Passive Fix	Geographic location of the vehicle, reusing host app's background GPS fixes to preserve battery.
GPS Accuracy Radius (m)	1 Hz	Horizontal uncertainty radius (68% confidence in metres) for inverse-variance weighting.
Speed & Heading (Bearing)	1 Hz	Speed (km/h) and direction (0–360°), essential for speed-adaptive scaling and geofence timing.
High-Res Timestamps	Per-event (ns / ms)	Elapsed realtime monotonic clock mapped to epoch milliseconds to prevent clock drift.

2. Deterministic Signal Processing (DSP) & Mathematical Calculations

Before running any AI, the phone's CPU executes deterministic mathematical transformations over a **2.0-second sliding window** (50% overlap, 200 samples/axis):

Processing Step	Formula / Transformation	Engineering Objective
1. Linear Resampling	$\text{LinearInterpolate}(t, 100\text{Hz})$	Normalizes jittery Android sensor ticks onto a fixed 100 Hz grid, preventing frequency distortion.
2. Gravity Removal	$a_{\text{dyn}} = a_{\text{total}} - g$	Subtracts constant 9.81 m/s² gravity vector so only dynamic road shocks remain.
3. Coordinate Rotation	$R_{\text{gravity.heading}} \cdot [aX, aY, aZ]^T$	Converts arbitrary phone orientation into vehicle frame: a_{vertical} , $a_{\text{longitudinal}}$, a_{lateral}
4. Total Shock Mag.	$a_{\text{mag}} = \sqrt{(aX^2 + aY^2 + aZ^2)}$	Rotation-invariant kinetic shock magnitude; stable even if phone shifts during riding.
5. Band-Pass Filter	Butterworth (0.5 – 30 Hz)	Strips low-frequency chassis drift (<0.5 Hz) and engine hum (>30 Hz), isolating 3–20 Hz pothole energy.
6. Self-Calibration	$z_{\text{norm}} = (a_{\text{vert}} - \mu_{\text{dev}}) / \sigma_{\text{dev}}$	Calculates rolling baseline noise floor (μ , σ) over smoothest 20% of driving to equalize soft vs. stiff vehicles.
7. Maximum Jerk	$\text{jerk}_{\text{max}} = \max(dz/dt)$	Rate of change of vertical acceleration (m/s³); speed-stable metric separating true potholes from bumps.
8. Windowed RMS	$\text{RMS} = \sqrt{(1/N) \sum z_i^2}$	Root Mean Square over 2-second window to gauge continuous surface roughness.
9. Cross-Axis Corr	$\text{Corr}(a_{\text{vertical}}, a_{\text{lateral}})$	Differentiates single-wheel pothole drop (roll asymmetry) from two-wheel speed bump hits (symmetrical pitch).
10. Threshold Gate	if $ z_{\text{norm}} < 3.0$ and $\text{RMS} < r0$: DROP	The Battery Saver: Hard rule-based gate dropping 99% of windows; model runs only ~5 times/minute.

3. On-Device AI Classification & Decision Engine (LiteRT / 1D-CNN)

Candidate windows surviving the threshold gate are processed locally by an INT8-quantised 1D-CNN (~120 KB) running on LiteRT:

Class Index & Name	Physical Waveform Signature & Characteristics	Action / Routing
Class 0: smooth_road	Minimal vertical variance, smooth laminar vibration floor.	Drop locally (no upload)
Class 1: rough_road	Sustained high-frequency vibration, elevated windowed RMS, no distinct crater.	Queue event (if $\tau \geq 0.8$)
Class 2: POTHOLE	Sharp drop-then-rebound vertical signature + high jerk + single-wheel roll asymmetry.	Queue event (if $\tau \geq 0.8$)
Class 3: speed_bump	Symmetrical upward pitch spike, dual-wheel compression, low roll correlation.	Drop locally (False Positive)
Class 4: rumble_strip / joint	Periodic multi-peak spikes (expansion joint, railway crossing, manhole cover).	Drop locally (False Positive)
Class 5: non_road_event	Phone drop off mount, pocket movement, door slam, sudden foot-braking.	Drop locally (False Positive)

4. Exact Information Sent to the Cloud Backend (Payload Contract)

When anomalies pass the AI confidence threshold ($\tau \geq 0.8$), the app batches them into gzipped JSON uploaded every 60s:

```
// POST /v1/events (Batched, Gzipped JSON)
{
  "sdk_version": "0.4.1",
  "device": {
    "id_hash": "sha256(install_id + salt)", // Salt-hashed device ID (No PII / GDPR-compliant)
    "model_class": "android_midrange", // Generalized device tier (reduces fingerprinting)
    "vehicle_class": "two_wheeler", // two_wheeler | car | auto | bus
    "calibration": { "noise_floor_z": 0.42, "samples": 60000 } // Baseline noise floor calibration
  },
  "events": [
    {
      "ts": 1786512345678, // Anomaly epoch timestamp in milliseconds
      "lat": 26.84671, "lon": 80.94623, // GPS coordinates of the detected defect
      "gps_accuracy_m": 6.4, // 68% confidence error radius (metres)
      "speed_kmph": 34.2, "heading_deg": 271.5, // Instantaneous speed and compass bearing
      "label": "pothole", // 6-class AI output (pothole | rough_road)
      "confidence": 0.87, // AI model softmax confidence probability
      "peak_z": 18.4, // Peak vertical acceleration (m/s² in vehicle frame)
      "jerk_max": 220.5, // Maximum shock jerk (m/s³)
      "rms_window": 4.9, // Windowed Root Mean Square roughness
      "duration_ms": 180 // Impact shock duration in milliseconds
    }
  ],
  "passes": [
    { "segment_id": "osm:way/1234", "count": 12 } // The Denominator: silent passes over road segment
  ]
}
```

Additional Server-Command Payloads (Triggered via Response ACK):

- **Mode 2 (Human Confirmation Vote):** { cmd_id: 'c_1b7d', answer: 'yes'|'no'|'not_sure', ts: 1786512999000, device_id_hash: '...' } — One-tap verification prompt displayed strictly when vehicle is stopped (speed < 3 km/h for > 5s).
- **Mode 3 (Gated Visual Proof):** 5–8 second 720p rear-camera video clip uploaded directly to pre-signed S3 URL only at confirmed candidate geofences (60–80m speed-adaptive radius) when phone is in a stable mount. Local video deleted immediately upon HTTP 200 and checksum verification.

5. Edge Performance, Privacy & Telemetry Budgets

Performance Metric	Hard Budget	Enforcement & Architectural Mechanism
Battery Consumption	< 2% / hour	Threshold gate drops 99% of windows; reuses host GPS stream; no camera in normal mode.
Mobile Data Transfer	< 1 MB / day	Batched gzipped JSON (~200 bytes/event); only uploads high-confidence positives ($\tau \geq 0.8$).
CPU & Memory Footprint	< 3% CPU / < 5 MB RAM	120 KB INT8 LiteRT model executed single-threaded via XNNPACK; circular in-memory buffer.
Offline Resilience	100% Offline-First	Events queue locally in Room SQLite database; WorkManager retries upload with exponential backoff.
Privacy (DPDP Act 2023)	Zero PII / Salted Hashes	No continuous GPS tracking (only discrete defect points); salted SHA-256 device hashing; instant erasure API.